

Assignment 4 Crafting a solution to a vulnerable network

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Introduction

The objective of this paper is to perform a rudimentary risk assessment of a generic smart home network consisting of an array of Internet of things (IoT) devices as well as other home devices. Part of this risk assessment is I will be constructing a network diagram meant to mitigate some of the risks identified here. IoT devices are increasingly prevalent throughout the average home and yet these devices also come with numerous vulnerabilities that put consumers personal information at grave risk. This paper will be divided into three main sections, identifying 5 devices and their vulnerabilities, a network design diagram that I am recommending to improve security, and last an in depth explanation of the proposed network and its improvements.

Artifact 1: Risk Assessment

The first part of this risk assessment is to identify several common home network connected devices, I will be primarily focused on IoT devices but will also include a smart phone to demonstrate the impacts of a consumer's phone being compromised. The five devices I will be assessing are, The Google Home Mini, Leeo Smart Alert, Phillips Hue Smart Bulb, Amazon Echo, and a generic smartphone. For each item I will be identifying information stored on the device, physical assets the item protects and would be compromised, threats, potential impacts, and finally a risk matrix score.

Google Home Mini

The first device is the Google Home mini, which functions as a smart speaker that can receive and respond to voice commands and control other smart home devices plus more. The Google home mini would store information such as voice commands, account credentials and information about the home Wi-Fi connection. This device can control other IoT devices meaning that it protects many other IoT devices like cameras, door locks, smart appliances and potentially many other IoT devices. Potential threats to this device include both internal and external, an authorized house member could exploit the device internally or a hacker wanting to gain unauthorized access could present an external threat. The impact of the Google Home Mini being compromised is very high, specifically because if this device is compromised it is highly likely that a threat would gain access or control over all or most of the other IoT devices on this network, creating what is known as a snowball effect. In addition to gaining control over other networked devices an attacker would also gain access to sensitive voice data and personal information (credentials) of the device owner. I would rate the risk of this device as high, mainly due to the wide impact if the device is compromised.

Leeo Smart Alert System

The second device I will discuss is the Leeo Smart Alert system, which is a fire and flood prevention alert system that notifies the user if a smoke, carbon monoxide, or water alarm is activated in the user's home. The Leeo smart alert system would store information such as app credentials and the status of home alert systems. The Leeo smart alert system protects other systems such as fire, water, and carbon monoxide detection systems. An internal attacker would disable or even gain some control over these systems for nefarious means. An example of an external attacker would be a cyber criminal that for example performs a DoS attack to disable the mentioned emergency alert systems. The impact of the Leeo smart system being compromised is that being unaware of emergency systems and compromising of the general safety of the home and the occupants of the home. For

this device I rated the risk as medium because it does have a potentially significant impact. However, the breach would be localized and limited to connected alert systems.

Phillips Hue Smart Bulb

The next device to be discussed is the Phillips Hue Smart bulb. This device is an IoT light system that can be controlled over a network interface. The Phillips Smart bulb would store information such as network pairing information and lighting system information. Should this device be compromised the only physical devices that would be compromised would be the light system itself. A threat to this device would be an external attack seeking to disrupt user's using the light system or potentially gain information about how the light system is used. The impact of this device being compromised is that an attacker could gain control of the automation of this light system, and they could learn information about the user's habits via how they use the light system and even potential network vulnerabilities. I rate the risk score of this device as medium because it is mostly limited to the direct impact on the lighting system.

Amazon Echo

The next device I have identified is the Amazon Echo, which is a smart speaker system that can respond to voice commands and also able to control some other IoT devices. The Amazon Echo would store information such as voice commands, account credentials, and information needed to compromise other IoT devices. If the Amazon Echo were to be compromised it would compromise other physical devices such as other IoT devices like door locks, cameras, and many other devices. Threats to this device include both internal such as house members misusing the device and external attackers looking to compromise the home network. The potential impacts of the Amazon echo being compromised are signification, an attacker could use the Amazon echo to make authorized purchases, gain personal information about the owner such as voice interactions, and even gain control of other smart home devices and their related home automation. For this device I give a risk score of high, due to the potential for remote controlling other devices and the large amount of sensitive data exposure.

Generic Smartphone

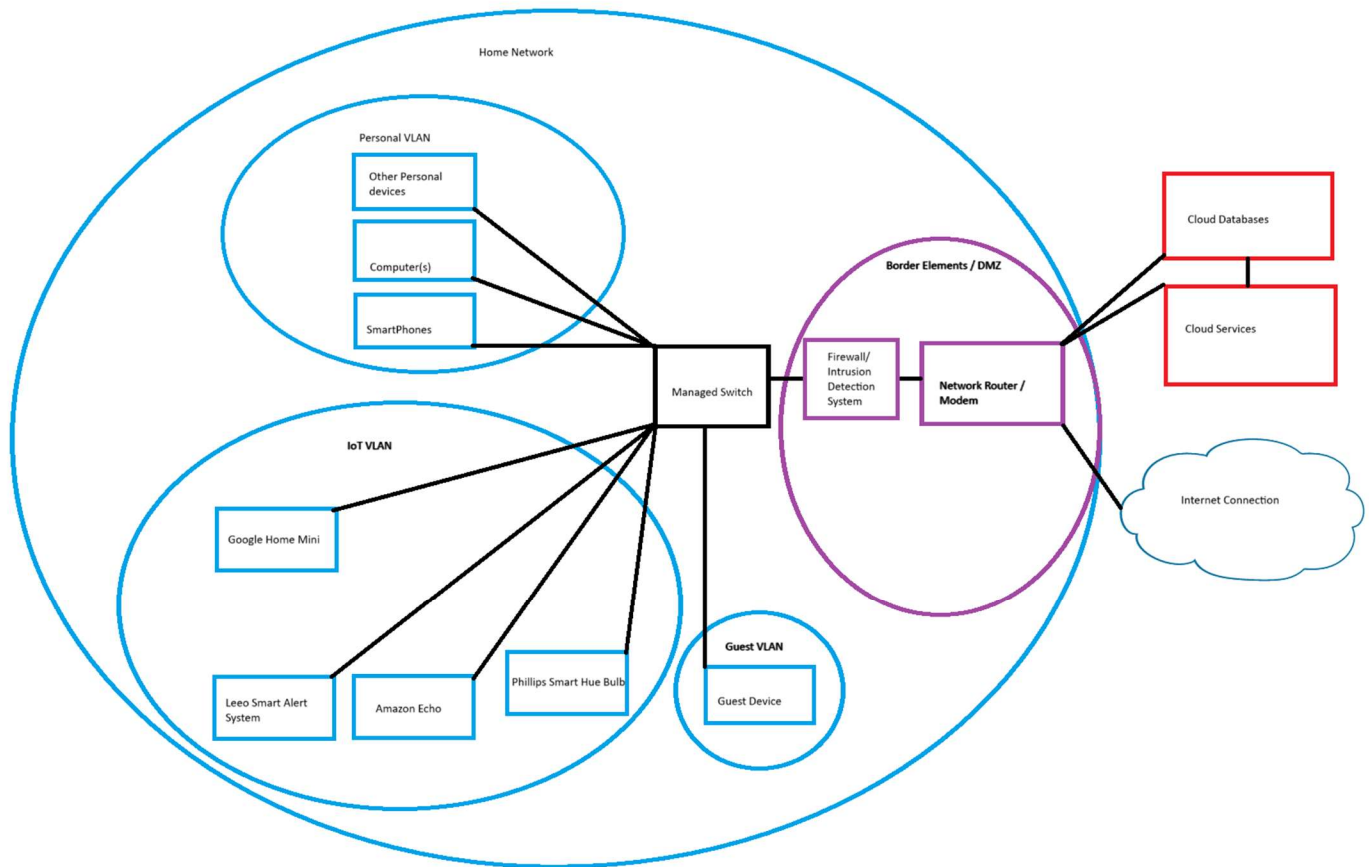
The last device that I will be discussing is a scenario where a homeowner's smart phone itself is compromised. In the modern era smartphones are connected to almost everything we use and contain nearly all of a person's personal data. The information that could be stored and thus accessed through a smart phone is immense and includes but is not limited to things such as personal accounts, banking details, passwords, authentication tokens, and direct control over nearly all IoT devices. Threats to a user's

smart phone include a array of threats, such as house hold members wanting to misuse the system, external attackers wanting to gain personal information or control over a user's various accounts, and even attackers wanting to gain control over a user's home automation systems. The external threat could be wanting to simply disrupt a user's control over their system or could be far more malicious and obtain personal information to compromise the user's identity. The impact from a smart phone being compromised is similarly extremely high, an attacker could gain access to financial accounts, assume control over nearly all other IoT devices on the network, and steal personal information that could be used to steal the user's identity. In addition to the immediate impact, compromising a smart phone is a perfect example of the Snowball effect where if this device is compromised the attacker is able to access many other devices. In this situation the snowball effect is so significant that an attacker would gain control over nearly all other devices. For this device I give a high-risk score, for several reasons. First the amount of personal information stored on a smartphone is immense, it would contain nearly all of a user's personal information. In addition to the personal information the extensive snowball effect would mean that an attacker would gain control over nearly the entire home network, thus compromising all the devices previously mentioned in this report.

Executive Summary

The devices discussed here are common devices that many homes now have. All these devices introduce significant vulnerabilities to home networks and also contain significant amounts of personal information. The security postures of companies that design these IoT devices need to be reviewed and adjusted to better contain the personal information that they use. Additionally, more steps need to be taken to better contain the snowball effect or spread of a breach should certain devices be breached. Devices such as smartphones, Amazon Echo, and Google home devices all would allow for a significant snowball affect should the initial device be compromised. Threats to these home networks include an array of attackers, including members of the household who are misusing devices, attackers attacking over the internet, and even attackers in the local neighborhood looking to physically compromise devices. The impact of many of these devices is very high and widespread, making the overall risk of these IoT devices very high. While IoT devices have significantly augmented our ability for home automation we must balance those features with increased security when designing new products.

Artifact 2: Network Diagram



Artifact 3: Network Design

The network re-design that I am proposing is comprised of a few significant changes. First, I would create separate VLANs to separate devices based on their usage and thus limit the impact of certain types of breaches. Specifically, I would establish a VLAN for IoT devices, a personal VLAN for members of the household, and a guest VLAN for temporary users. The purpose of creating these VLANs is to separate devices based on data usage and limit exposure of devices, limit access to personal data, and allow for different levels of security for devices (Four Strategic Principles of Network Security Design, 2020). The network itself would be managed by a central security hub or router which would manage access to the network. This system would make use of public and private key encryption such as. The administrative access to the network would be secured using a strong MFA system. Access to IoT devices and the IoT VLAN would be secured using a RSA based encryption. To protect against the imposter key threat, the system will use digital certificates to verify trusted devices, this key will be frequently changed to prevent or limit

attacks (Kay, 2018). To secure communication TLS 1.3 will be used to encrypt all communication. A zero-trust model will be implemented to require verification for every access attempt. These protocols are being used because TLS 1.3 prevents data interception, and a zero-trust method minimizes insider threats.

Executive Summary

In summary, smart home networks need to adjust their security posture in order to mitigate vulnerabilities introduced by IoT devices. The number of devices in consumer homes is increasing drastically and often these devices are not designed with a strong security posture. However, even a home network can be adjusted to help mitigate many of these networks. Creating separate VLAN networks goes a long way to limit and mitigate some of the threats created by IoT devices (Four Strategic Principles of Network Security Design, 2020). This separation protects personal data and other network devices from being exposed should one device be breached (Four Strategic Principles of Network Security Design, 2020). It is also important to incorporate public and private key encryption schemes in order to harden the security of all devices on the network. In order to address what is known as an imposter key networks need to utilize a zero-trust system and require an authentication system which also checks digital signatures regularly (Kay, 2018). Last, modern encryption protocols like TLS 1.3 need to be utilized to protect all communication that is being transmitted out or into the network. IoT devices provide an array of new features and convenience to modern users, however these devices need to be safeguarded by equally modern security methods, protocols, and software systems such as firewalls and encryption methods.

Bibliography

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